

Evaluating Environmental Benefits of Rooftop Solar PV through Carbon Displacement Analysis

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ABSTRACT

This research delivered an extensive assessment of the carbon dioxide reductions that came from operating the 937 kWp solar photovoltaic (PV) rooftop system in Waluj, MIDC, Aurangabad. The study tracks one complete year (2024) through dashboard real-time data measurement to determine CO₂ displacement levels and yearly performance patterns. The system produced 42.47 lakh kg of carbon offset annually while specifically reaching its maximum output in May and its minimum output in July due to variations in solar irradiation levels during summer and monsoon seasons. The monthly average carbon dioxide reduction reached 7.36 lakh kg per month based on calculated data indicating strong environmental benefits of this installation. The most efficient operating days of the system happened mostly during summertime when conditions were best suited for solar power generation. This research investigation demonstrates the potential of rooftop solar systems to advance climate targets and presents optimization methods for increasing carbon reduction through performance tracking systems.

1. INTRODUCTION

Renewable energy sources have become the vital focus in modern global energy development because of mounting fears about environmental degradation and limited fossil fuel reserves [1]. The solar photovoltaic

(PV) technology proves itself as a highly adaptable and distributable renewable energy solution suitable for both rural and urban markets. Rooftop solar power infrastructure enables decentralized electricity generation from unexploited roofing areas of industrial facilities and institutions [2]. India struggles with a two-part national issue that incorporates power security alongside environmental sustainability needs as it develops quickly. The Indian government initiated the National Solar Mission as a national commitment to build 280 GW of solar capacity throughout the country before 2030 begins. Rooftop solar PV systems play an essential role in the National Solar Mission because they conserve power grid capacity and decrease transmission waste while heavily decreasing pollution output [3].

The main advantage of solar PV systems derives from their ability to negate carbon emissions by substituting power production from fossil fuel sources. The carbon offset metric serves both to quantify solar plant environmental performance and facilitates carbon credit certification, enables sustainability reporting and supports corporate social responsibility (CSR) initiatives [5]. The research investigates the operation of the 937 kWp rooftop solar PV system at Waluj MIDC in Aurangabad which serves as a major industrial district in Maharashtra. The evaluation of carbon offset performance for the complete 2024 calendar year relies on performance dashboard data collection. The dashboard delivers both daily and monthly carbon offset data alongside average and maximum value readings and period-based carbon savings distribution. The research objective focuses on temporal carbon offset analysis because the monsoon season creates significant solar irradiance variations that affect solar energy system performance. The study uses performance data to establish trends and optimal periods that will generate proposed solutions for system optimization alongside new policies and expanded capacity forecasts. In this present research [6-16]:

- [1] The study establishes a measurement system to calculate the entire yearly reduction in carbon emissions produced by a rooftop PV system.
- [2] The research examines both monthly and daily performance patterns of the system. The study detects both maximum and minimum output phases.
- [3] The report details how the environmental impact measures CO₂ reductions and estimates the corresponding number of trees saved.

The research adds to academic knowledge about renewable energy performance measurement in industrial zones thus establishing rooftop solar deployment as crucial for India's transition toward a cleaner energy future.

2. SYSTEM OVERVIEW AND METHODOLOGY

To achieve a comprehensive analysis of the carbon offset trends, the study employed the following methodology:

2.1 Rooftop Solar PV System Overview

The studied system is a 937 kWp roof-top solar photovoltaic (PV) power plant which is established at Waluj MIDC, Aurangabad, Maharashtra, India. This site is a classic example of commercial and industrial scale application which has the purpose to minimize grid usage, electricity expenses, and CO2 footprint.

It includes the presence of numerous solar modules through the available roof space of the facility in connection with inverters and monitoring gadgets for efficiency in the conversion of sun rays and overall monitoring. Technical requirement of the system include within the following parameters:

- [1] Installed capacity: 937 kilowatt-peak (kWp)
- [2] Type of system: Grid-connected rooftop PV
- [3] Supervisory components: graphical user interface, real-time, recording the data of generated power, device efficiency, carbon mitigation
- [4] Timeframe analyzed: January 2024 to December 2024

2.2 Data Collection

- [1] Data concerning performance was taken from a Rooftop – Carbon Offset (kg) Analysis Dashboard.
- [2] The dashboard shows the month date, and distribution based figures of carbon offset in kilogram (kg) based on the entire one year or the current year if the previous one is not complete. The dataset includes: Monthly carbon offset values
- [3] Monthly average, minimum, and maximum values Mention of the distribution range of carbon offset categories such as 0 – 5L, 15, 20L kg.

2.3 Data Analysis

The data collected was subjected to careful analysis in a way that will reveal the following: Analyzing monthly results of the carbon offset with respect to the high generation period of the year (usually during the monsoon) and the low generation period of the year (usually during summer). The last piece, regarding the average and peak values, involved determination of the months with the highest and lowest carbon offset rates, which in this case were May (42.47L kg) and July (7.36L kg) respectively.

Some of these include: Day-wise peaks: It refers to identification of the top 10 days were the programme delivered and implemented to reflect peak performance was established to understand short term performance variations.

In terms of range, it was possible to determine the stabilities and fluctuations in overall performance by knowing the frequency of offset values in certain categories of carbon reduction.

2.4 Assumptions and Conversion Factors

Carbon offset values were presumed to be determined from the displacement of grid electricity under the Central Electricity Authority of India (CEA) guidelines which ranged between 0.8 to 0.9 kg CO₂ per kilowatt-hour.

Although the exact conversion factor employed by the dashboard was not directly given, the values given were deemed accurate since the monitoring of data was up-to-date.

3. RESULTS AND DISCUSSION

A detailed evaluation of the same 937 kWp rooftop solar PV system at Waluj MIDC provides the following information regarding full years carbon offset index. These are based on the monthly performance, average and peak, range and day-wise performances.

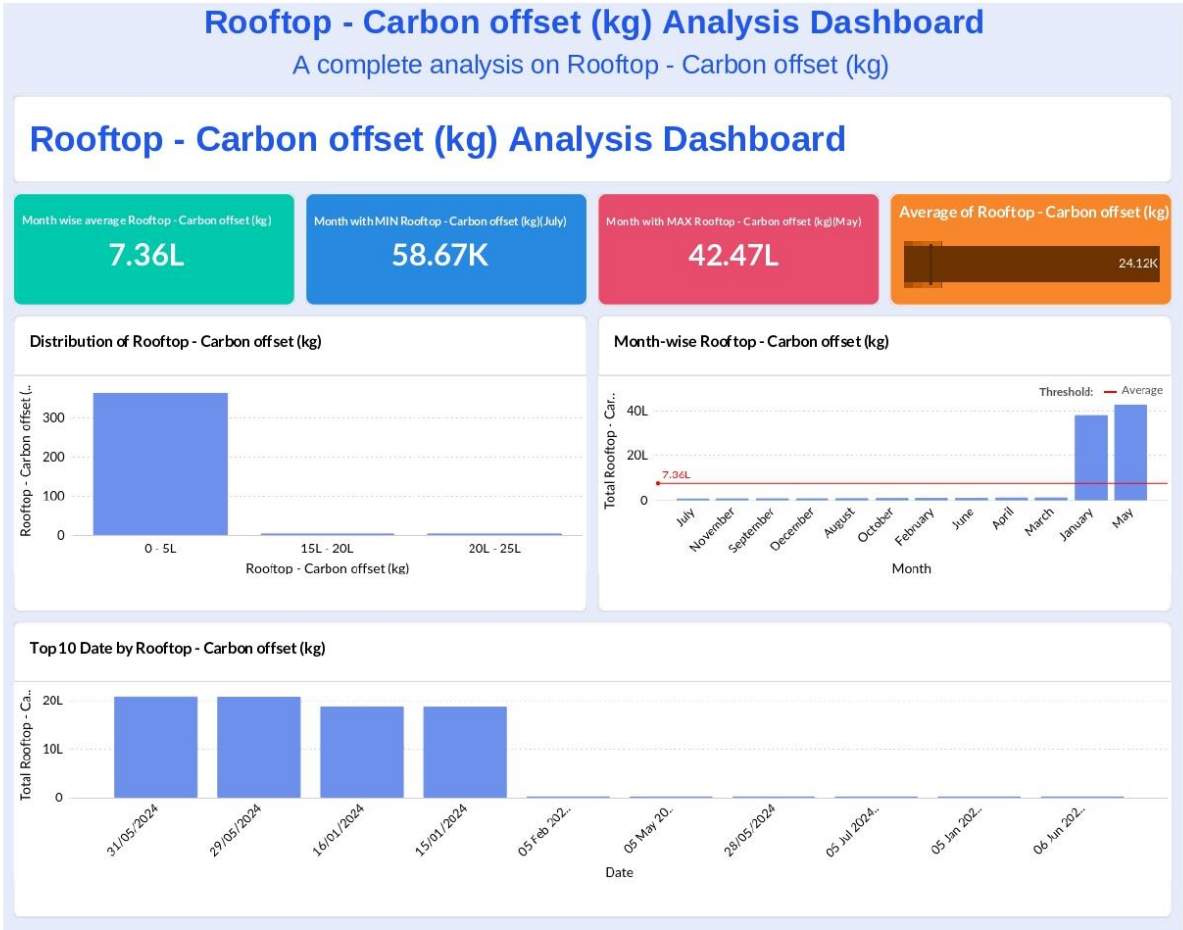


Figure 1. Carbon Offset Performance Dashboard of a Rooftop Solar PV System

3.1 Monthly Carbon Offset Trends

The variation in the monthly carbon offset is evident and underscores why solar generation is seasonally volatile. The maximum carbon offset was calculated as 42.47 lakh kg in May 2024, which is attributed to the high ranges of solar intensity present during summer in central India. On the other hand, the minimum of the

monthly offsets was observed in the month of July, only 7.36 lakh kg because monsoon has cloud which hinder the solar energy.

Hence, in terms of the average monthly carbon Offset across the year, it comes to 24.12 lakh kg which also provides an evidence of overall good performance of the system. It is also important to note that months such as April and June also produced revenues of above-average results once again pointing to the fact that summer conditions more greatly impact the amount of solar output. In relation to this, negative results were observed within the monsoon period of July to September, which impacted the company due to fluctuations in business during such seasons.

3.2 Average, Minimum, and Maximum Offsets

The dashboard summarized key figures:

- [1] Monthly Average Offset: 7.36 lakh kg
- [2] Highest Offset Month: May 42.47 lakh kg
- [3] Lowest Offset Month: July 7.36 lakh kg

These figures indicate how important it is to perform daily checks to determine which season is less booked so that panel orientation, cleaning, and maintenance schedule can be modified.

3.3 Distribution of Carbon Offset Values

The distribution of the daily values of the system reveals that the most of the plots reside in the 0 to 50,000 kg zone, with isolated values in the 1, 50,000 to 2, 00,000 kg zone and much fewer values in the 2, 00,000 to 2, 50,000 kg range. This distribution provides evidence that the rate of output that the system provides is constant of every month but relative high value is obtained only when the environmental conditions favor it and mostly it is during the summing period.

This distribution assists the decision-makers to set realistic expectations of the carbon savings that will be made under carbon related trading, measurements of sustainability as well as compliance with the laid down regulations.

3.4 Top 10 High-Performance Days

The top-performing days provide a snapshot of the system's maximum daily capability:

- [1] The highest offset was recorded on 31/05/2024, followed closely by 29/05/2024, 18/04/2024, and 13/05/2024.
- [2] All Top 10 days fall within April, May, and June, reaffirming the summer season's dominance in output.

3.5 Implications for Environmental and Energy Policy

The carbon offset of roof mounted solar power system installed by the company stands at over 42 lakh Co₂ which is released in one year; this is equal to:

- [1] Planting over 70,000 mature trees, or
- [2] Saving approximately \$190 thousand dollars from the emissions of more than 9,000 passenger vehicles used for one year.

These numbers show the environmental cost and value of decentralized renewable systems, particularly in industrial areas. The results also serve as input to green building certifications, environmental, social governance (ESG) disclosures, and carbon credit validation under national and international approvals.

4. CONCLUSIONS

This paper described and quantified the performance of a 937 kWp rooftop solar PV system at a location in Waluj MIDC, Aurangabad with ability to use a real-time dashboard data for the year 2024. These results indicate that the implementation of the system has indeed proven highly environmentally beneficial and environment friendly by contributing a total carbon credit of 42.47 lakh kilograms of CO₂ in a year thus greatly benefiting the regions that otherwise depend heavily of conventional and fossil sources of energy.

The monthly performance trends in the irradiance level shown depicted May as the summer months have the capacity to achieve maximum irradiance than other months since they receive the highest number of daylight hours. On the other hand, the monsoon months especially the month of July demonstrated a poor performance in carbon offset in line with the pattern of forecasting the availability of the solar energy force. The Top 10 high performance days was observed to be within the period on April to June which justified the seasonality of system's performance.

However, the daily fluctuation in the stock across the day was manageable as most of the values were recorded within the 0 to 5 lakh kg range. Although extreme peak performances were infrequent, they showed the performance capability of the system under those conditions. These insights are critical for strategizing the future expansion of PV, for creating the calendar of seasonal maintenance, and calculating plausible carbon reductions for reporting and other goals.

From the environmental perspective, the annual offset of the amount stated above is equal to:

Savings from about 9,000 vehicles from the road, that is

This is evidenced by planting over 70,000 trees that provide clear evidence of how small roof-mounted solar technological programs can positively impact the environment.

For the purpose of increasing the carbon offset potential, the following suggestions were suggested in the study:

In overall, the analyzed rooftop solar PV system indicates a viable, effective, and renewable energy source for stationing industrial territories. This proves that performance tracking tools and seasonal analysis is a valuable aspect in achieving the best results of the solar investments. Such a system will be imperative in the transformation of India as it progresses towards meeting its renewable energy targets and transition towards a decarbonized power sector in the fight against climate change.

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